* Vector differential operator ∇

The vector differential operator denoted by ∇ (read as del or nabla) is defined by

$$\nabla = \frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}.$$

This operator has no meaning on itself but assumes specific meaning depending on how it operates on a scalar or vector point function.

* Gradient of a scalar point function

Let $\phi(x,y,z)$ be a scalar point function defined and differentiable at each point (x,y,z) in a certain region of space. Then the gradient of ϕ, written as $\nabla\phi$ or grad ϕ is defined by

$$\text{grad}\,\phi = \nabla\phi = \frac{\partial\phi}{\partial x}\hat{i} + \frac{\partial\phi}{\partial y}\hat{j} + \frac{\partial\phi}{\partial z}\hat{k}$$

We see that $\nabla\phi$ is a vector function. Thus, the gradient of scalar point function is a vector point function.

- Note:

1. $|\nabla\phi| = \sqrt{\left(\frac{\partial\phi}{\partial x}\right)^2 + \left(\frac{\partial\phi}{\partial y}\right)^2 + \left(\frac{\partial\phi}{\partial z}\right)^2}$ - magnitude of grad ϕ
2. If ϕ is a scalar point function, then $\frac{\partial\phi}{\partial x}, \frac{\partial\phi}{\partial y}, \frac{\partial\phi}{\partial z}$ are called components of grad ϕ.
3. Gradient of ϕ is the vector whose direction is the direction in which ϕ increases the fastest and whose magnitude is the derivative of f in that direction.

- Physical meaning of gradient.

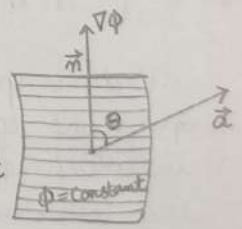
The gradient is the change in quantity with respect to distance (space) or say when we move along the length (simple case).

Like temperature gradient. Suppose you had a rod of some length and you put one end of it above a gas burner, while you hold the other end. Now after sometime you will feel the rod is getting hot. But it will be less hotter at your hand held end and more at burner side end. The temperature will keep on increasing as you go to the burner side. The temperature is increasing along the length of the rod.

- **Properties of gradient**

1. $d\phi = \nabla\phi \cdot d\vec{r}$ where $d\vec{r} = dx\,\hat{i} + dy\,\hat{j} + dz\,\hat{k}$.
2. The necessary and sufficient condition for a scalar point function $\phi(x,y,z)$ to be a constant is that $\nabla\phi = 0$.
3. For any two scalar functions ϕ and ψ,
   (i) $\nabla(c_1\phi \pm c_2\psi) = c_1\nabla\phi \pm c_2\nabla\psi$ where c_1 and c_2 are constants.
   (ii) $\nabla(\phi\psi) = \phi\nabla\psi + \psi\nabla\phi$
   (iii) $\nabla\left(\frac{\phi}{\psi}\right) = \frac{\psi\nabla\phi - \phi\nabla\psi}{\psi^2}$ $(\psi \neq 0)$

- **Directional derivative : Normal derivative :**



Let $\phi(x,y,z)$ be a scalar point function and $\vec{a}$ be a vector inclined at an angle θ to the direction of $\nabla\phi$. Then the component of $\nabla\phi$ along $\vec{a}$, namely $\nabla\phi \cdot \hat{a}$ (where $\hat{a}$ is the unit vector along $\vec{a}$) is called the directional derivative of ϕ along $\vec{a}$. This is denoted by $\frac{\partial\phi}{\partial\vec{a}}$.

Thus, $\frac{\partial\phi}{\partial\vec{a}} = \nabla\phi \cdot \hat{a} = |\nabla\phi|\cos\theta$.

In particular the directional derivative of ϕ along $\vec{n}$ is

$$\frac{\partial \phi}{\partial \vec{n}} = \nabla\phi \cdot \hat{n} = |\nabla\phi|$$

This is called the normal derivative of ϕ.

If $\theta = 0$, $\cos\theta = 1$ which is maximum value of $\cos\theta$.

$$\therefore \quad \max \frac{\partial \phi}{\partial \vec{a}} = |\nabla\phi| \cos 0 = |\nabla\phi| = \frac{\partial \phi}{\partial \vec{n}}$$

Thus the directional derivative $\frac{\partial \phi}{\partial \vec{a}}$ is maximum when $\vec{a}$ is directed along $\vec{n}$ and the maximum is equal to the normal derivative. This means that ϕ varies most rapidly along $\nabla\phi$ and $|\nabla\phi|$ gives the maximum rate of variation.

Note :

(i) The unit normal vector $\hat{n}$ directed along $\nabla\phi$ is given by

$$\hat{n} = \frac{\nabla\phi}{|\nabla\phi|}.$$

* Geometrical meaning of grad ϕ. (Before directional derivative)

- Level surfaces : Let $\phi(P)$ define a scalar field over a region R. The point P satisfying an equation of the form $\phi(P) = c$ (c is an arbitrary constant) constitute a family of curved lines in two dimensional space and a family of surfaces in three dimensional space. The curved lines given by $\phi(P) = c$ in two dimensional space are called contour lines, where as the surfaces $\phi(P) = c$ in three dimensional space are called as level surfaces. At each point on this surface the function ϕ has the same value as at P. This level surface is also known as equi potential surface.

Eg: For two dimensional field $\phi(P) = x^2+y^2$, the contour lines are circles $x^2+y^2=c^2$. For three dimensional field $\phi(P) = x^2+y^2+z^2$, the level surfaces are spheres $x^2+y^2+z^2=c^2$ (c is an arbitrary constant).

- Geometrical meaning :

We know that $\phi(x,y,z) = c$ represent a level surface where c is a constant. Let $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ be the position vector of any point $P(x,y,z)$ on the surface. Then $d\vec{r} = dx\hat{i} + dy\hat{j} + dz\hat{k}$ lies in the plane tangent to the surface at P.

$$\text{grad}\,\phi \cdot d\vec{r} = \left(\frac{\partial\phi}{\partial x}\hat{i} + \frac{\partial\phi}{\partial y}\hat{j} + \frac{\partial\phi}{\partial z}\hat{k}\right).(dx\hat{i} + dy\hat{j} + dz\hat{k})$$

$$= \frac{\partial\phi}{\partial x}dx + \frac{\partial\phi}{\partial y}dy + \frac{\partial\phi}{\partial z}dz$$

i.e., $\nabla\phi \cdot d\vec{r} = d\phi$. But $d\phi = 0$ $(\because \phi = c)$

$\therefore \nabla\phi \cdot d\vec{r} = 0$.

Hence $\nabla\phi$ is orthogonal to $d\vec{r}$ and therefore to the surface. Therefore grad ϕ is along the normal to the surface $\phi(x,y,z) = c$ at $P(x,y,z)$.

- Note : The angle between the normal's to the surfaces is given by $\cos\theta = \hat{n}_1 \cdot \hat{n}_2$

* Problems :

1. If $\phi(x,y,z) = xy^2z^3 - x^3y^2z$, then find $\nabla\phi$ and $|\nabla\phi|$ at $(1,-1,1)$.

Sol: Given $\phi(x,y,z) = xy^2z^3 - x^3y^2z$

$$\frac{\partial\phi}{\partial x} = y^2z^3 - 3x^2y^2z \quad ; \quad \frac{\partial\phi}{\partial y} = 2xyz^3 - 2x^3yz \quad ;$$

$$\frac{\partial\phi}{\partial z} = 3xy^2z^2 - x^3y^2$$

$$\nabla\phi = \frac{\partial\phi}{\partial x}\hat{i} + \frac{\partial\phi}{\partial y}\hat{j} + \frac{\partial\phi}{\partial z}\hat{k}$$

$$= (y^2z^3 - 3x^2y^2z)\hat{i} + (2xyz^3 - 2x^3yz)\hat{j} + (3xy^2z^2 - x^3y^2)\hat{k}$$

$$\nabla\phi\big|_{(1,-1,1)} = (1-3)\hat{i} + (-2+2)\hat{j} + (3-1)\hat{k} = -2\hat{i} + 2\hat{k}$$

$$|\nabla\phi| = \sqrt{(-2)^2 + 0^2 + 2^2} = \sqrt{8} = 2\sqrt{2}$$

2. Find the directional derivative of $\phi(x,y,z) = x^2yz + xz^2$ at the point $(1,-2,1)$ in the direction of $2\hat{i} - \hat{j} + 2\hat{k}$.

Sol

$$\phi(x,y,z) = x^2yz + xz^2$$

$$\nabla\phi = \frac{\partial\phi}{\partial x}\hat{i} + \frac{\partial\phi}{\partial y}\hat{j} + \frac{\partial\phi}{\partial z}\hat{k}$$

$$= (2xyz + z^2)\hat{i} + (x^2z)\hat{j} + (x^2y + 2xz)\hat{k}$$

$$\nabla\phi\Big|_{(1,-2,1)} = (-4+1)\hat{i} + \hat{j} + (-2+2)\hat{k} = -3\hat{i} + \hat{j}$$

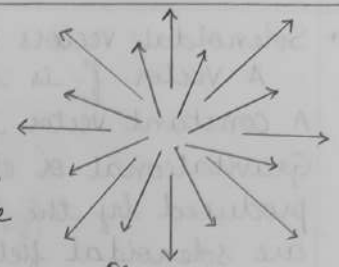
Given $\vec{a} = 2\hat{i} - \hat{j} + 2\hat{k}$

Directional derivative of ϕ is $\nabla\phi.\hat{a} = (-3\hat{i}+\hat{j}).(2\hat{i}-\hat{j}+2\hat{k})/\sqrt{4+1+4}$

$$= \frac{-6-1}{3} = \frac{-7}{3}$$

3.

* Divergence of a vector field



The concept of gradient describes the rate of change of a scalar field. Similarly divergence and curl describe the rate of change of vector field.
The divergence of a vector field is a scalar field that tells us at each point the extent to which the field explodes or diverges from that point.

- **Definition : Divergence of a vector function**

Let $\vec{f}(x,y,z) = f_1(x,y,z)\hat{i} + f_2(x,y,z)\hat{j} + f_3(x,y,z)\hat{k}$ be a continuously differentiable vector point function. Then divergence of a vector point function denoted by $\text{div}\,\vec{f}$ or $\nabla\cdot\vec{f}$ is defined as

$$\text{div}\,\vec{f} = \nabla\cdot\vec{f} = \left(\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}\right)\cdot\left(f_1\hat{i} + f_2\hat{j} + f_3\hat{k}\right)$$

$$\text{div}\,\vec{f} = \nabla\cdot\vec{f} = \frac{\partial f_1}{\partial x} + \frac{\partial f_2}{\partial y} + \frac{\partial f_3}{\partial z}$$

$\therefore$ Divergence of vector point function is a scalar point function.

- **Note** : 1. Divergence of a constant vector is identically equal to zero.
2. The symbol $\nabla\cdot\vec{f}$ does not represent the dot product of ∇ and $\vec{f}$ in the sense of dot product of two vectors, because the symbol ∇ is just a differential operator and not a vector.
3. $\nabla\cdot\vec{f} \neq \vec{f}\cdot\nabla$

- **Physical significance** :

If $\vec{v}$ is the velocity of a moving fluid at a point P at time t, then div $\vec{v}$ represents the rate at which the fluid flows out of a unit volume enclosing the point P (i.e., rate of expansion per unit volume under the flow of fluid).
Similar meaning is attached, when $\vec{v}$ is any other physical quantity which is a vector field.

* **Solenoidal Vectors :**

A vector $\vec{f}$ is said to be solenoidal if $\text{div}\,\vec{f} = 0$.

A constant vector is a trivial example of a solenoidal vector. Gravitational or electrostatic fields which are fields of force produced by the inverse square law of attraction or repulsion are solenoidal fields.

* **Properties of divergence :**

1. For any scalar constants a and b and any vector fields $\vec{F}$ and $\vec{G}$, we have

$$\nabla.(a\vec{F} \pm b\vec{G}) = a\nabla.\vec{F} \pm b\nabla.\vec{G}$$

2. For any scalar field ϕ and any vector field $\vec{F}$,

$$\text{div}(\phi\vec{F}) = \phi\,\text{div}\,\vec{F} + (\nabla\phi).\vec{F}$$

i.e.,

$$\nabla.(\phi\vec{F}) = \phi(\nabla.\vec{F}) + (\nabla\phi).\vec{F}$$

* **Problems :**

1. If $\vec{f} = x^2y\,\hat{i} - 2xz\,\hat{j} + 2yz\,\hat{k}$ then find $\text{div}\,\vec{f}$.

Sol:

$$\text{div}\,\vec{f} = \nabla.\vec{f} = \frac{\partial f_1}{\partial x} + \frac{\partial f_2}{\partial y} + \frac{\partial f_3}{\partial z}$$

$$= \frac{\partial}{\partial x}(x^2y) + \frac{\partial}{\partial y}(-2xz) + \frac{\partial}{\partial z}(2yz)$$

$$= 2xy + 0 + 2y = 2y(x+1)$$

2. If $\vec{f} = 3xy\,\hat{i} + x^2z\,\hat{j} - y^2e^{2z}\,\hat{k}$, then find $\nabla.\vec{f}$ at $(1,2,0)$.

Sol:

$$\text{div}\,\vec{f} = \nabla.\vec{f} = \frac{\partial f_1}{\partial x} + \frac{\partial f_2}{\partial y} + \frac{\partial f_3}{\partial z}$$

$$= \frac{\partial}{\partial x}(3xy) + \frac{\partial}{\partial y}(x^2z) + \frac{\partial}{\partial z}(-y^2e^{2z})$$

$$= 3y + 0 - 2y^2e^{2z}$$

At $(1,2,0)$, $\nabla.\vec{f} = 6 - 2(4) = -2$

3. If $\vec{f} = \frac{x\hat{i}+y\hat{j}}{x+y}$ then find div $\vec{f}$.

Sol: Given $\vec{f} = \frac{x}{x+y}\hat{i} + \frac{y}{x+y}\hat{j}$

$$\text{div } \vec{f} = \frac{\partial f_1}{\partial x} + \frac{\partial f_2}{\partial y} = \frac{\partial}{\partial x}\left(\frac{x}{x+y}\right) + \frac{\partial}{\partial y}\left(\frac{y}{x+y}\right)$$

$$= \frac{(x+y)1 - x(1)}{(x+y)^2} + \frac{(x+y)1 - y(1)}{(x+y)^2}$$

$$= \frac{y}{(x+y)^2} + \frac{x}{(x+y)^2} = \frac{1}{(x+y)}$$

4. If $\phi(x,y,z) = 2x^3y^2z^4$, then find div(grad ϕ)

Sol: $$\text{grad } \phi = \nabla\phi = \frac{\partial\phi}{\partial x}\hat{i} + \frac{\partial\phi}{\partial y}\hat{j} + \frac{\partial\phi}{\partial z}\hat{k}$$

$$= 6x^2y^2z^4\,\hat{i} + 4x^3yz^4\,\hat{j} + 8x^3y^2z^3\,\hat{k}$$

$$\text{div(grad } \phi) = \frac{\partial f_1}{\partial x} + \frac{\partial f_2}{\partial y} + \frac{\partial f_3}{\partial z}$$

$$= \frac{\partial}{\partial x}(6x^2y^2z^4) + \frac{\partial}{\partial y}(4x^3yz^4) + \frac{\partial}{\partial z}(8x^3y^2z^3)$$

$$= 12xy^2z^4 + 4x^3z^4 + 24x^3y^2z^2$$

5. If $\vec{f} = 3xy\hat{i} + 20yz^2\hat{j} - 15xz\hat{k}$ and $\phi = y^2 - xz$, then find div($\phi\vec{f}$)

Sol: $$\phi\vec{f} = (y^2 - xz)(3xy\hat{i} + 20yz^2\hat{j} - 15xz\hat{k})$$

$$= (3xy^3 - 3x^2yz)\hat{i} + (20y^3z^2 - 20xyz^3)\hat{j} - (15xy^2z - 15x^2z^2)\hat{k}$$

$$\text{div}(\phi\vec{f}) = \frac{\partial}{\partial x}(3xy^3 - 3x^2yz) + \frac{\partial}{\partial y}(20y^3z^2 - 20xyz^3) - \frac{\partial}{\partial z}(15xy^2z - 15x^2z^2)$$

$$= 3y^3 - 6xyz + 60y^2z^2 - 20xz^3 - 15xy^2 + 30x^2z$$

6. Show that the vector function $\vec{f} = 2xyz\hat{i} + (xy - y^2z)\hat{j} + (x^2 - zx)\hat{k}$ is solenoidal.

Sol

$$\text{div}\,\vec{f} = \frac{\partial}{\partial x}(2xyz) + \frac{\partial}{\partial y}(xy - y^2z) + \frac{\partial}{\partial z}(x^2 - zx)$$

$$= 2yz + x - 2yz - x$$

$$= 0$$

$\therefore \vec{f}$ is solenoidal.

7. If $\vec{f} = (ax + 3y + 4z)\hat{i} + (x - 2y + 3z)\hat{j} + (3x + 2y - z)\hat{k}$ is solenoidal vector field, then find the value of a.

Sol If $\vec{f}$ is solenoidal, then $\text{div}\,\vec{f} = 0$.

$$\therefore \text{div}\,\vec{f} = \frac{\partial}{\partial x}(ax + 3y + 4z) + \frac{\partial}{\partial y}(x - 2y + 3z) + \frac{\partial}{\partial z}(3x + 2y - z) = 0$$

$$\Rightarrow a - 2 - 1 = 0 \Rightarrow a = 3$$

8. If $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ and $r = |\vec{r}|$, then show that $\text{div}(r^n\vec{r}) = (n+3)r^n$.

Sol Given $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$

$$r = |\vec{r}| = \sqrt{x^2 + y^2 + z^2}$$

$$\Rightarrow r^2 = x^2 + y^2 + z^2$$

$$2r\frac{\partial r}{\partial x} = 2x \Rightarrow \frac{\partial r}{\partial x} = \frac{x}{r}$$

Similarly, $\frac{\partial r}{\partial y} = \frac{y}{r}$, $\frac{\partial r}{\partial z} = \frac{z}{r}$

Now, $r^n\vec{r} = r^n(x\hat{i} + y\hat{j} + z\hat{k})$

$$= r^n x\hat{i} + r^n y\hat{j} + r^n z\hat{k}$$

$$\text{div}(r^n\vec{r}) = \sum\left[\frac{\partial}{\partial x}(r^n x)\right]$$

$$= \sum\left[r^n + n r^{n-1}\cdot\frac{\partial r}{\partial x}\cdot x\right] = \sum r^n + \sum\left(x\cdot n r^{n-1}\cdot\frac{x}{r}\right)$$

$$= 3r^n + n r^{n-2}(x^2 + y^2 + z^2) = (n+3)r^n.$$

9. If $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ and $r = |\vec{r}|$, then show that $\frac{\vec{r}}{r^3}$ is solenoidal.

Soln Given $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$

$$r = \sqrt{x^2+y^2+z^2} \Rightarrow r^2 = x^2+y^2+z^2$$

$$\frac{\partial r}{\partial x} = \frac{x}{r} \; ; \; \frac{\partial r}{\partial y} = \frac{y}{r} \; ; \; \frac{\partial r}{\partial z} = \frac{z}{r}$$

Now, $\frac{\vec{r}}{r^3} = \frac{x}{r^3}\hat{i} + \frac{y}{r^3}\hat{j} + \frac{z}{r^3}\hat{k}$

$$\operatorname{div}\left(\frac{\vec{r}}{r^3}\right) = \sum \frac{\partial}{\partial x}\left(\frac{x}{r^3}\right) = \sum \frac{r^3(1) - x.3r^2\frac{\partial r}{\partial x}}{(r^3)^2}$$

$$= \sum \frac{r^3 - 3xr^2.\frac{x}{r}}{r^6} = \sum \frac{r^3 - 3x^2r}{r^6}$$

$$= \sum \frac{r^3}{r^6} - \sum \frac{3x^2r}{r^6}$$

$$= \frac{3}{r^3} - \frac{3}{r^5}(x^2+y^2+z^2)$$

$$= \frac{3}{r^3} - \frac{3}{r^5}(r^2) = 0$$

$\therefore \frac{\vec{r}}{r^3}$ is solenoidal.

* The Laplacian of a scalar field :

In electrostatics, the gradient of the electric potential is a scalar multiple of the electric field intensity, and the divergence of the electric field intensity is related to the charge density. For this and other reasons, it is convenient to introduce a single operator that is the composite of two operators grad and div. This operator is called the Laplacian.

The Laplacian of a scalar field ϕ is defined to be div(grad ϕ). Note that grad ϕ is a vector field and the divergence of grad ϕ is a scalar field. Hence the Laplacian of scalar field ϕ is a scalar field. In del notation it is represented as $\nabla\cdot(\nabla\phi)$ and for simplicity it is frequently written as $\nabla^2\phi$.

$$\text{Laplacian}(\phi) = \nabla^2\phi = \nabla\cdot\nabla\phi = \left(\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}\right)\cdot\left(\frac{\partial\phi}{\partial x}\hat{i} + \frac{\partial\phi}{\partial y}\hat{j} + \frac{\partial\phi}{\partial z}\hat{k}\right)$$

$$= \frac{\partial}{\partial x}\left(\frac{\partial\phi}{\partial x}\right) + \frac{\partial}{\partial y}\left(\frac{\partial\phi}{\partial y}\right) + \frac{\partial}{\partial z}\left(\frac{\partial\phi}{\partial z}\right)$$

$$\nabla^2\phi = \frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} + \frac{\partial^2\phi}{\partial z^2}$$

The above expression is often written in the form

$$\nabla^2\phi = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right)\phi$$

so that ∇^2 is the differential operator given by

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

This differential operator is called the Laplacian operator.

• Harmonic functions :

A function satisfying Laplace equation is called a Harmonic function. Thus a scalar field ϕ is a harmonic function whenever $\nabla^2\phi = 0$. This equation is called Laplace equation.

1. If $\vec{x} = x\hat{i} + y\hat{j} + z\hat{k}$ and $r = |\vec{x}|$, then prove that
$\nabla^2 f(r) = f''(r) + \frac{2}{r} f'(r)$.

Sol. $\vec{x} = x\hat{i} + y\hat{j} + z\hat{k}$

$r = \sqrt{x^2+y^2+z^2} \Rightarrow r^2 = x^2+y^2+z^2$

$\frac{\partial r}{\partial x} = \frac{x}{r}$; $\frac{\partial r}{\partial y} = \frac{y}{r}$; $\frac{\partial r}{\partial z} = \frac{z}{r}$

$$\nabla^2 f(r) = \frac{\partial^2}{\partial x^2} f(r) + \frac{\partial^2}{\partial y^2} f(r) + \frac{\partial^2}{\partial z^2} f(r)$$

$$= \sum \frac{\partial^2}{\partial x^2} f(r) = \sum \frac{\partial}{\partial x}\left(\frac{\partial}{\partial x} f(r)\right)$$

$$= \sum \frac{\partial}{\partial x}\left(f'(r)\frac{\partial r}{\partial x}\right) = \sum \frac{\partial}{\partial x}\left(f'(r)\cdot\frac{x}{r}\right)$$

$$= \sum \left[\frac{f'(r)}{r} + x\left\{\frac{r f''(r)\frac{\partial r}{\partial x} - f'(r)\frac{\partial r}{\partial x}}{r^2}\right\}\right]$$

$$= \sum \left[\frac{f'(r)}{r} + \frac{x r f''(r)\cdot\frac{x}{r} - x f'(r)\cdot\frac{x}{r}}{r^2}\right]$$

$$= \sum \left[\frac{f'(r)}{r} + \frac{x^2}{r^2} f''(r) - \frac{x^2}{r^3} f'(r)\right]$$

$$= \frac{3f'(r)}{r} + \frac{f''(r)}{r^2}(x^2+y^2+z^2) - \frac{f'(r)}{r^3}(x^2+y^2+z^2)$$

$$= \frac{3f'(r)}{r} + \frac{f''(r)}{r^2}\cdot r^2 - \frac{f'(r)}{r^3}\cdot r^2$$

$$= \frac{3f'(r)}{r} + f''(r) - \frac{f'(r)}{r}$$

$$= f''(r) + \frac{2}{r} f'(r)$$

2. If $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ and $r = |\vec{r}|$ then prove that $\nabla^2(r^{n+1}) = (n+1)(n+2)r^{n-1}$

Sol. Given $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$

$$r = |\vec{r}| \quad ; \quad r^2 = x^2 + y^2 + z^2$$

$$\frac{\partial r}{\partial x} = \frac{x}{r} \quad ; \quad \frac{\partial r}{\partial y} = \frac{y}{r} \quad ; \quad \frac{\partial r}{\partial z} = \frac{z}{r}$$

$$\nabla^2 r^{n+1} = \sum \frac{\partial^2}{\partial x^2}(r^{n+1}) = \sum \frac{\partial}{\partial x}\left(\frac{\partial}{\partial x} r^{n+1}\right)$$

$$= \sum \frac{\partial}{\partial x}\left((n+1) r^n \cdot \frac{\partial r}{\partial x}\right)$$

$$= (n+1) \sum \frac{\partial}{\partial x}\left(r^n \frac{x}{r}\right) = (n+1) \sum \frac{\partial}{\partial x}(r^{n-1} x)$$

$$= (n+1) \sum \left[r^{n-1} + x(n-1) r^{n-2} \frac{\partial r}{\partial x} \right]$$

$$= (n+1) \sum \left[r^{n-1} + x(n-1) r^{n-2} \cdot \frac{x}{r} \right]$$

$$= (n+1)\left[3r^{n-1} + (n-1) r^{n-3} (x^2 + y^2 + z^2) \right]$$

$$= (n+1)\left[3r^{n-1} + (n-1) r^{n-1} \right]$$

$$= (n+1) r^{n-1} (3 + n - 1)$$

$$\nabla^2 r^{n+1} = (n+1)(n+2) r^{n-1}$$

3. If $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ and $r = |\vec{r}|$ then prove that $\operatorname{div}(r^n \vec{r}) = (n+3) r^n$ and hence show that $\nabla^2(r^n \vec{r}) = n(n+3) r^{n-2} \vec{r}$.

Sol. We have already proved that $\operatorname{div}(r^n \vec{r}) = (n+3) r^n$.

$$\nabla^2(r^n \vec{r}) = \nabla\left[\nabla \cdot (r^n \vec{r})\right] = \nabla\left[(n+3) r^n\right]$$

$$= (n+3) \nabla r^n = (n+3)\left[\frac{\partial}{\partial x} r^n \hat{i} + \frac{\partial}{\partial y} r^n \hat{j} + \frac{\partial}{\partial z} r^n \hat{k} \right]$$

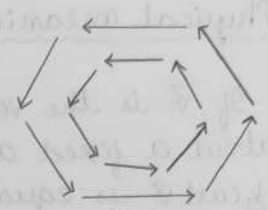
$$= (n+3)\left[n (r)^{n-1} \frac{\partial r}{\partial x} \hat{i} + n r^{n-1} \frac{\partial r}{\partial y} \hat{j} + n r^{n-1} \frac{\partial r}{\partial z} \hat{k} \right]$$

$$= n(n+3)\left[r^{n-1} \cdot \frac{x}{r} \hat{i} + r^{n-1} \frac{y}{r} \hat{j} + r^{n-1} \cdot \frac{z}{r} \hat{k} \right]$$

$$= n(n+3) r^{n-2} \left[x\hat{i} + y\hat{j} + z\hat{k} \right]$$

$$= n(n+3) r^{n-2} \vec{r}$$

* Curl of a vector field :



The curl of a vector field is a vector field that gives us at each point an indication of how the field swirls in the vicinity of that point.

Let $\vec{F}(x,y,z)$ be a vector point function which is continuously differentiable. When this vector point function $\vec{F}$ is operated vectorially by the operator ∇, we get a vector quantity known as curl of the vector, denoted as curl $\vec{F}$ or $\nabla \times \vec{F}$.

Let $\vec{F} = F_1\hat{i} + F_2\hat{j} + F_3\hat{k}$

$$\text{curl } \vec{F} = \nabla \times \vec{F} = \left(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z}\right)\hat{i} + \left(\frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x}\right)\hat{j} + \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y}\right)\hat{k}$$

For computational purposes, the above expression is written in the following determinant form

$$\nabla \times \vec{F} = \text{curl } \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{vmatrix}$$

* Note :

1. If c is a scalar then curl $(c\vec{F}) = c$ curl $\vec{F}$
2. Curl of a vector field is a vector quantity.
3. The symbol $\nabla \times \vec{F}$ does not represent the cross product of ∇ and $\vec{F}$ in the sense of cross product of two vectors, because ∇ is just a differential operator and not a vector.

* Irrotational Vector :

A vector $\vec{F}$ is said to be irrotational if curl $\vec{F} = 0$.

A constant vector field is a trivial example of an irrotational vector.

* Physical meaning of curl :

If $\vec{V}$ is the velocity of a particle in a rigid body rotating about a fixed axis with a uniform angular velocity, then $\frac{1}{2}$ curl $\vec{V}$ is equal to the angular velocity of the body.

Also, curl $\vec{F}$ is associated physically with rotation or spin of fluid. Hence it is also written as rot $\vec{F}$.

* Properties :

1. If $\vec{V}$ is a constant vector, $\nabla \times \vec{V} = 0$.
2. If $\vec{F}$ and $\vec{G}$ are two vector functions, then
$$\nabla \times (\vec{F} + \vec{G}) = (\nabla \times \vec{F}) + (\nabla \times \vec{G})$$
3. For any scalar field ϕ, curl $\nabla\phi = 0$ i.e., $\nabla \times (\nabla\phi) = 0$.

* Conservative vectors :

A vector field $\vec{F}$ is said to be conservative vector if there exists a scalar function ϕ such that $\vec{F} = \nabla\phi$, then ϕ is called a scalar potential of $\vec{F}$.

We know that $\nabla \times \nabla\phi = 0$. From this it follows that if $\vec{F}$ is a conservative vector then curl $\vec{F} = 0$. Thus every conservative vector is an irrotational vector.

* Problems :